affirmoTM tutorial 3: settings

Each setting in settings.yaml, and how to change them

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This tutorial starts from a fresh analysis folder with the example input files (see tutorial 1):

```
library(affirmoTM)
```

```
tm_example_inputs()
```

1 The settings file

`inputs/settings.yaml` holds the run settings, one per line, with comments:

```
cat(readLines("inputs/settings.yaml"), sep = "\n")
```

```
# affirmoTM run settings.

# Stratifier columns in the transition matrices (and the cohort, entry states, and optionally costs and
# utilities). Every column of a matrix that is not `from` and not listed here is read as a state.
# The age-group column must be called `agegrp`: it is the only stratifier that changes as patients age.
# Any other name (for example `age_band`) is read as a fixed stratifier, so patients would keep their
# entry age group for the whole horizon. Use `strata: []` for matrices without stratifiers.
strata: [agegrp, sex, mm_cluster]

# Seed for the random draws (entry states and yearly transitions), so a run can be repeated exactly.
seed: 2026

# Time horizon in cycles (years), counted from cycle 0 (2013). Everyone stops at the end of cycle
# horizon - 1, so patients who enter later are followed for fewer years.
horizon: 10

# A patient who would pass the oldest age group (for example reach 110 when the oldest group is
# 105-109) dies in that year, in this death state.
```

```
age_limit_death_state: CurrentDeath

# Annual discount rate for costs and QALYs (0.03 = 3%). Discounting runs from cycle 0 (2013) for
# everyone: a year in cycle t counts 1 / (1 + rate)^t.
discount_rate: 0.03

# Number of times the whole simulation is run. Run r uses seed + r - 1, so every run differs and the
# whole set can be repeated exactly. Each run gives one point on the cost-effectiveness plane.
n_runs: 100

# Events to count in the results. Each year a patient spends in one of the listed states counts as
# one event. Deaths are counted automatically from the death states.
events:
  ischaemic_stroke:    [AIS, AIS_AB, PB_AIS, AIS_Death, AIS_AB_Death]
  haemorrhagic_stroke: [AHS, AHS_AB, PB_AHS, AHS_Death, AHS_AB_Death]
  major_bleed:         [AB, AIS_AB, AHS_AB, PS_AB, AB_Death, AIS_AB_Death, AHS_AB_Death]

# Currency symbol used in printed results and plots.
currency: "€"

# Willingness-to-pay thresholds per QALY, drawn as lines on the cost-effectiveness plane. Points
# below a line are cost-effective at that threshold. Use `wtp: []` for no lines.
wtp: [20000, 30000, 50000]
```

Setting	Example	What it does
`strata`	`[agegrp, sex, mm_cluster]`	The stratifier columns of the input files. Every other matrix column, apart from `from`, is a state. `[]` for none
`seed`	2026	The starting seed. Run r uses `seed + r - 1`, so results can be repeated exactly
`horizon`	10	The number of yearly cycles, counted from cycle 0 for everyone
`age_limit_death_state`	`CurrentDeath`	The death state of patients who would pass the oldest age group
`discount_rate`	`0.03`	The yearly discount rate for costs and QALYs, as a proportion (`0.03` for 3%). A year in cycle t counts 1 / (1 + rate)^t
`n_runs`	100	The number of runs; each run is one point on the CE plane
`events`	lists of states	The states counted as each event in the results
`currency`	`"€"`	The currency symbol in the printed results and the plot
`wtp`	`[20000, 30000, 50000]`	Willingness-to-pay thresholds per QALY, drawn as lines on the CE plane. `[]` for none

`tm_read_settings()` reads the file and checks every setting. A wrong type or value stops with a message naming the setting, and a name the package does not know, such as a typo, gives a warning.

In R, the settings are a list:

```r
settings <- tm_read_settings()
str(settings)
```

```
List of 9
 $ strata               : chr [1:3] "agegrp" "sex" "mm_cluster"
 $ seed                 : int 2026
 $ horizon              : int 10
 $ age_limit_death_state: chr "CurrentDeath"
 $ discount_rate        : num 0.03
 $ n_runs               : int 100
 $ events               :List of 3
```

```
  ..$ ischaemic_stroke   : chr [1:5] "AIS" "AIS_AB" "PB_AIS" "AIS_Death" ...
  ..$ haemorrhagic_stroke: chr [1:5] "AHS" "AHS_AB" "PB_AHS" "AHS_Death" ...
  ..$ major_bleed        : chr [1:7] "AB" "AIS_AB" "AHS_AB" "PS_AB" ...
 $ currency              : chr "€"
 $ wtp                   : int [1:3] 20000 30000 50000
```

2 Changing settings

There are two ways:

- **Edit `inputs/settings.yaml`** and run `tm_run()` again. This is the way to change the analysis itself.
- **Change the list in R** and pass it to `tm_run()`, which is handy for trying things out; the file stays as it is.

The examples below change the list in R, with 20 runs each to keep them quick. Every call to `tm_run()` overwrites `outputs/`.

```
base <- tm_read_settings()
base$n_runs <- 20
base_results <- tm_run(base)
```

2.1 Discount rate

```
no_discounting <- base
no_discounting$discount_rate <- 0
no_discounting_results <- tm_run(no_discounting)
```

2.2 Time horizon

The example cohort enters between cycles 0 and 9. With a 5-year horizon, the patients who would enter in cycles 5 to 9 fall outside it, and `tm_run()` leaves them out with a warning:

```
five_years <- base
five_years$horizon <- 5
five_years_results <- tm_run(five_years)
```

```
Warning: inputs/cohort.csv: 1343 patients enter in cycle 5 or later, after the
horizon of 5 cycles, and are left out (ids 3658, 3659, 3660, 3661, 3662, 3663, 3664,
3665, 3666, 3667 and 1333 more).
```

`intervention_cost.csv` lists years 1 to 9, which covers a 5-year horizon too: later years are allowed and not used.

2.3 Side by side

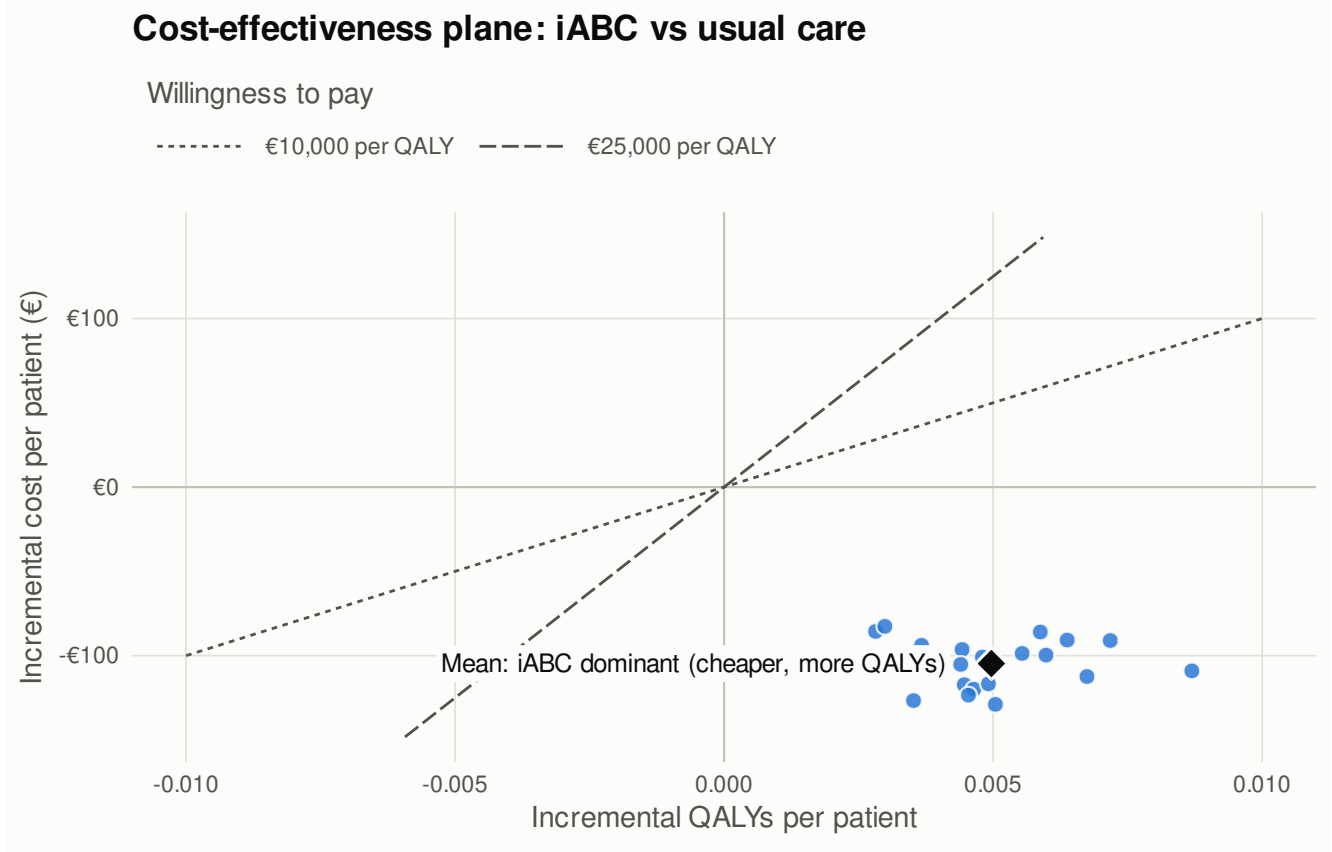
```
rbind(cbind(setting = "3% discount, 10 years", base_results$comparison),
      cbind(setting = "no discounting, 10 years", no_discounting_results$comparison),
      cbind(setting = "3% discount, 5 years", five_years_results$comparison)
)[, .(setting, delta_cost, delta_qaly, icer, quadrant)]
```

```
                    setting delta_cost  delta_qaly      icer quadrant
                     <char>      <num>       <num>     <num>   <char>
1:    3% discount, 10 years -104.67869 0.004967325 -21073.45 dominant
2: no discounting, 10 years -126.25450 0.005941800 -21248.53 dominant
3:     3% discount, 5 years  -38.71584 0.001815603 -21323.96 dominant
```

2.4 Willingness-to-pay lines

`wtp` only changes the lines on the CE plane:

```
other_lines <- base
other_lines$wtp <- c(10000, 25000)
plot(tm_run(other_lines))
```



2.5 Seed and number of runs

The same seed always gives the same results; another seed gives another set of runs:

```
two_runs <- base
two_runs$n_runs <- 2
first <- tm_run(two_runs)$comparison
again <- tm_run(two_runs)$comparison
two_runs$seed <- 1
other_seed <- tm_run(two_runs)$comparison
rbind(cbind(run = "seed 2026", first), cbind(run = "seed 2026 again", again),
      cbind(run = "seed 1", other_seed))[, .(run, delta_cost, delta_qaly)]
```

```
              run delta_cost  delta_qaly
            <char>      <num>       <num>
1:       seed 2026  -102.4714 0.003580796
2: seed 2026 again  -102.4714 0.003580796
3:          seed 1  -118.5145 0.004084200
```

More runs give more points on the CE plane and a steadier mean. Each run of the 5,000 example patients takes about a quarter of a second.